

#### X-ray and Electron Analytical Methods (19 L)

X-ray fluorescence analysis (XRF), X-ray absorption analysis (XAS), Electron probe microanalysis (EPMA), Energy dispersive spectroscopy (EDS), Wavelength dispersive spectroscopy (WDS), Electron energy loss spectroscopy (EELS), Secondary ion mass spectroscopy (SIMS), Auger electron spectroscopy (AES), X-ray photoelectron spectroscopy, Raman spectroscopy.

#### Physico-chemical Methods (20 L)

Spectrochemical Analysis: Atomic emission spectrography. Atomic absorption spectrography. Spectrophotometers and colorimeters.

Electro-analytical Techniques: Polarography, Electrogravimetry, Conductometry and Potentiometry.

Chromatographic Analysis: Gas chromatography.

Determination of non-metals by combustion and vacuum fusion analysis.

### 4. READINGS

#### 4.1 TEXTBOOKS::

1. L.E. Murr: Electron and Ion Microscopy and Microanalysis - Principles and Applications, Marcel Dekker.
2. C.R. Brundle and A.D. Baker: Electron Spectroscopy, Vol.1-3, Academic.
3. D.A.Skoog et al.: Principles of Instrumental Analysis, Brooks Cole
4. H.H. Willard et.al.: Instrumental Methods of Analysis, CBS.

#### 4.2 \*REFERENCE BOOKS::

1. B.K.Sharma: Instrumental Methods of Chemical Analysis, Goel.
2. Materials Characterization Techniques, (eds.) G. Sridhar et.al. NML, Jamshedpur.
3. A.I. Vogel: Text book of Quantitative Inorganic Analysis, ELBS Longma